

Every Coin Counts

Counting coins on a textured pad with classical image processing

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89.6%

recall

294 of 328 coins found

88.3%

precision

294 of 333 detections were coins

39 / 34

false / missed

FP non-coins · FN missed coins

12/26

exact counts

only 7 with no mistake at all

The problem

People count coins at a glance; a computer sees only pixels. Our phone photos add three traps: a white pad full of folds and seams, dark or motion-blurred shots, and buttons the size of a coin mixed in.

Question Can fixed classical rules, with no training, count only the coins under these conditions?

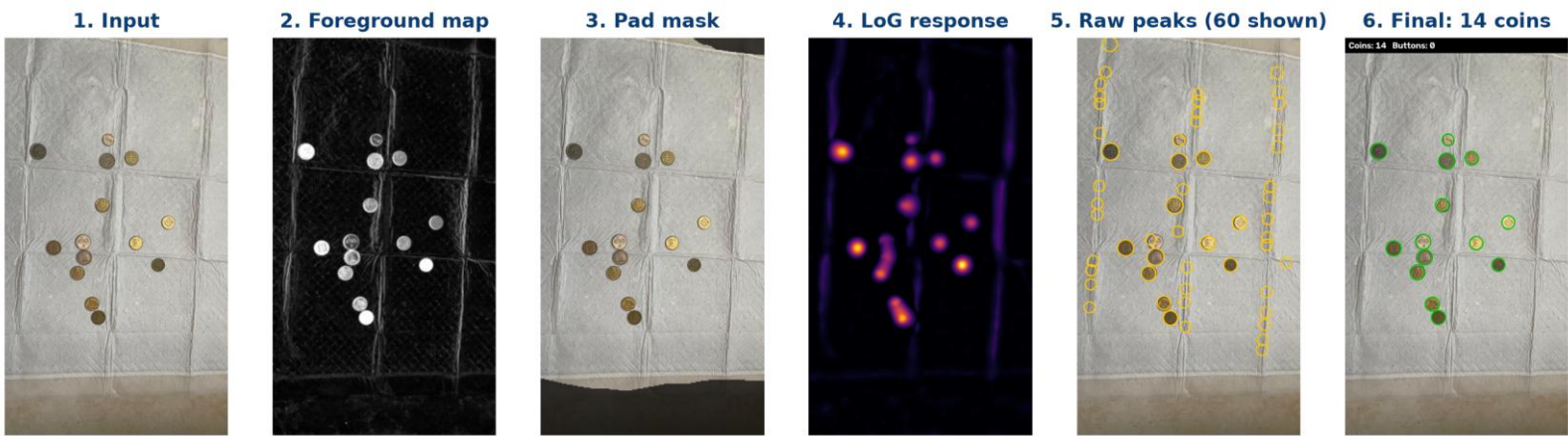
Data 26 photos · 328 coins · 12 buttons, hand-counted and split into clean, noisy (dark / blurred) and damaged (piles, cut coins, buttons). Then every circle the program drew was checked: coin (TP), not a coin (FP) or a coin without a circle (FN).

The pipeline



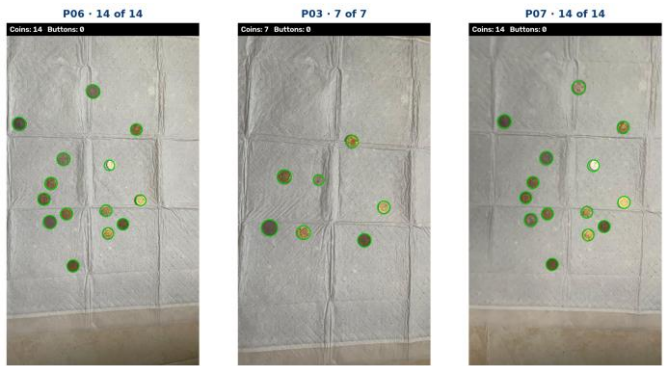
- **Background:** a huge median filter models pad and light; each pixel gets a Lab “difference from background” score.
- **Blobs:** multi-scale Laplacian of Gaussian finds round, coin-sized peaks.
- **Folds:** a Hessian test rejects ridges; a grayscale opening removes thin seams.
- **Touching coins:** Hough circles split what LoG merges.
- **Buttons:** black-hat reveals sewing holes; blue or pink objects are never coins.

From photo to count



Input → foreground map → pad mask → LoG response → raw peaks (mostly folds) → 14 coins after verification.

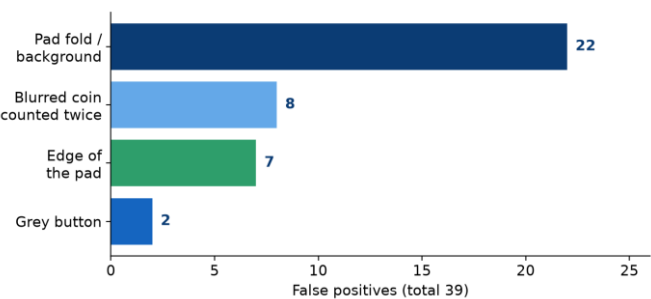
Results



Three clean photos with no mistake.

Clean photos: all 98 coins found, 1 false detection.
Dark photos add false coins; piles and the frame hide real ones.

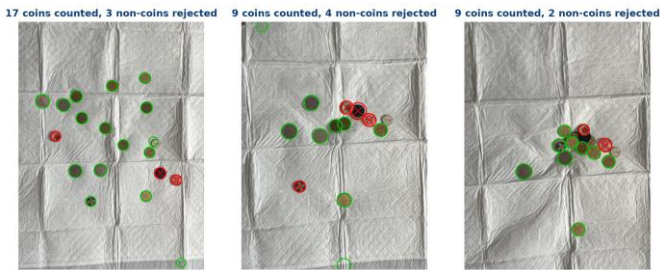
What was mistaken for a coin



Causes of the 39 false positives.

12 photos got the exact total, but in 5 of them a false coin and a missed coin cancelled out.

Round, but not a coin



Red X = button, not counted.

9 of 12 buttons rejected by colour or sewing holes (black-hat); 2 grey buttons passed as coins.

Conclusion

Classical tools alone found 89.6% of the coins with 88.3% precision and rejected 9 of 12 buttons. Checking every circle, not just the total, showed where and why the machine fails.

Limitations

Motion blur splits a coin into several blobs (22 counted vs 12). Dense piles hide coins, coins cut by the frame are half-circles, and a grey button without visible holes looks like a coin.

Next steps

Merge detections along the blur direction, drop the white hem from the pad mask, fit partial circles at the image border, and use rim colour to catch grey buttons.